Amit Roy

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Education

Purdue Univeristy **West Lafayette, Indiana, USA**

PhD Student in Computer Science | MSc in Computer Science *August 2022 - May 2028 (Expected)*

Research Focus: LLM Reasoning via Post-Training, Natural Language Processing, Applied Machine Learning

CGPA: 3.90 out of 4.00 | (Offered Ross Fellowship)

Univeristy of Dhaka **Dhaka, Bangladesh**

B.Sc in Computer Science and Engineering | Undergraduate Curriculum *January 2016 - August 2020*

CGPA: 3.96 out of 4.00 (Class Rank: 1st out of 65 students)

Thesis Topic: Utility-based Graph Mining

Professional Experience

Purdue Computer Science **West Lafayette, Indiana, USA**

PhD Student - LLM Personalization, LLM Reasoning, Personalized Anomaly Detection *August 2022 - Present*

Curently, working on personalized anomaly detection in social media and personalized response generation from conversation data using LLM post-training approaches. Previously, worked on *Graph Reasoning with transformers* and *Graph Anomaly Detection* project with autoencoders, energy-based models, funded by **Sony, Belgium**.

Amazon Inc. **Sunnyvale, California, USA**

Applied Scientist Intern - Prime Video Personalization & Discovery Team *May 2026 - August 2026*

Proposed criterion-based Group-Relative Policy Optimization (GRPO) algorithm for finetuning LLMs for personalized re-ranking problem of retrieved videos from a user query and personalized watch history in Amazon Prime Video platform, achieving 88-90% NDCG@5 score, trained on Movielens dataset for 10000 queries while ranking 20 candidate movies in each query with 10 movies in the user's watch history.

Amazon Inc. **Bellevue, Washington, USA**

Applied Scientist Intern - Finance Technology (FinTech) Team *May 2025 - August 2025*

Contributed on Amazon project *Minerva*, a platform for Natural Language Question Answering (NLQA) from uploaded documents, developing a query-spreadsheet and subquery-column matching for relevant spreadsheet retrieval and LLM Agents with code generation and execution capabilities achieving 80%-84% accuracy for NLQA from multiple spreadsheets | *Project Report*

Futurewei Technologies **Santa Clara, California, USA**

ML Algorithm Intern - Dynamic Text-Attributed Graph Learning with LLMs *May 2024 - August 2024*

Worked as an ML algorithm intern and developed an LLM-driven knowledge distillation framework for future link prediction and edge classification tasks for dynamic text-attributed graphs. | Advisors: Dr. Masood S. Mortazavi, Dr. Ning Yan

Artificial Intelligence and Cybernetics Lab, Independent University Bangladesh **Dhaka, Bangladesh**

Research Assistant - Machine Learning on Graphs *February 2020 - June 2022*

Developed GNN-based models for spatial-temporal traffic forecasting, node classification, and graph classification tasks supported by the ICT Division, Government of Bangladesh. | Advisors: Dr. Amin Ahsan Ali and Dr. AKM Mahbubur Rahman

Tiger IT Bangladesh Ltd. **Dhaka, Bangladesh**

Software Engineer - ML Engineer for Applied Computer Vision Projects *December 2020 - August 2022*

Worked on developing ML systems for computer vision projects e.g. automated number plate recognition, object detection from real-time video data using yolov3 and face reenactment with a first-order-motion model. | Team Lead: Shakur Shams Mullick

Research Projects

Personalized Anomaly Detection in Social Media with LLMs via Reinforcement Learning **September 2026**

***Amit Roy**, Bharat Bhargava*

Ongoing Project | *[Paper]*

Personalized Video Re-ranking via Reinforcement Learning with Criterion-Level Credit Assignment **August 2026**

***Amit Roy**, Bharat Bhargava, Jingru Cheng, Phanideep Gampa, Sheikh Sarwar, Chaosheng Dong, Ying Chen, Haiyang Zhang*

Internship Project @ Amazon Prime Video and Personalized Discovery | *[Paper]*

Zero-shot Persona Emulation from Conversational Data **May 2026**

***Amit Roy**, Bharat Bhargava*

Under Review *[Paper]*

Transformers Can Learn Connectivity on Some Graphs but Not Others **October 2025**

***Amit Roy**, Abulhair Saparov*

Under Review *[Paper]* *[Code]*

Natural Language-based Question Answering from Multiple Spreadsheets via RAG and code-based LLM Agents **August 2025**
***Amit Roy**, Zishan Muzeeb, Piyush Paritosh*
Internship Project @ Minerva, Amazon Fintech | *[Paper]*

LLM-driven Knowledge Distillation for Dynamic Text-Attributed Graphs **March 2025**
***Amit Roy**, Ning Yan, Masood S. Mortazavi*
Artificial Intelligence for Time Series Analysis (AI4TS) Workshop at AAAI 2025 *[Paper] [Code]*

GAD-NR: Graph Anomaly Detection via Neighborhood Reconstruction **March 2024**
***Amit Roy**, Juan Shu, Jia Li, Carl Yang, Olivier Elshocht, Jeroen Smeets, Pan Li*
The 17th ACM International Conference on Web Search and Data Mining (WSDM-2024)*[Paper] [Code] [PyGOD]*[***180+ citations***]

GAD-EBM: Graph Anomaly Detection using Energy-Based Models **December 2023**
***Amit Roy**, Juan Shu, Olivier Elshocht, Jeroen Smeets, Ruqi Zhang, Pan Li*
New Frontiers in Graph Learning (GLFRONTIERS), NeurIPS Workshop 2023 *[Paper]* [*Code*]

Before Joining Purdue CS:

Unified Spatio-Temporal Modeling for Traffic Forecasting using GNN **July 2021**
Amit Roy, Kashob Kumar Roy*, Amin Ahsan Ali, M Ashraful Amin and A K M Mahbubur Rahman*
The International Joint Conference on Neural Networks (IJCNN-2021) *[Paper] [Code]*

SST-GNN: Simplified Spatio-temporal Traffic forecasting model using Graph Neural Network **May 2021**
Amit Roy, Kashob Kumar Roy*, Amin Ahsan Ali, M Ashraful Amin and A K M Mahbubur Rahman*
The 25th Pacific-Asia conference on Knowledge Discovery and Data Mining (PAKDD-2021) *[Paper] [Code]*

Node Embedding using Mutual Information and Self-Supervision based Bi-level Aggregation **July 2021**
Kashob Kumar Roy, **Amit Roy***, Amin Ahsan Ali, M Ashraful Amin and A K M Mahbubur Rahman*
The International Joint Conference on Neural Networks (IJCNN-2021) *[Paper] [Code]*

Structure-Aware Hierarchical Graph Pooling using Information Bottleneck **July 2021**
Kashob Kumar Roy, **Amit Roy***, Amin Ahsan Ali, M Ashraful Amin, and A K M Mahbubur Rahman*
The International Joint Conference on Neural Networks (IJCNN-2021) *[Paper] [Code]*

Mining High Utility Subgraphs **December 2021**
*Md. Tanvir Alam, **Amit Roy**, Chowdhury Farhan Ahmed, Md. Ashraful Islam, Carson K. Leung*
UDML 2021: 4th Workshop on Utility Driven Mining and Learning @ ICDM 2021 *[Paper] [Code]*

iGest Recognito : A Mobile App for Sign Language-based Communication **February 2018**
***Amit Roy**, Tauhid Tanjim, Saif Mahmud, Md. Mofijul Islam, Sarker Tanvir Ahmed Rumee*
International conference on Emerging technologies in data Mining and Information Security (IEMIS), Kolkata, India *[Poster] [Code]*

Awarded champion in the *Apps in Life* category at National IT Competition on 2018 [*Award Link*]

**** indicates Equal Contribution***

Academic Projects

Impact of Large Language Models in Database Management Systems *August 2023 - December 2023*
Performed evalution on the performance of LLMs e.g. GPT4 for DBMS tasks e.g. text-to-SQL, query disambiguation. [Presentation]

Investigating Knowlege Graph Completion with Pre-trained Language model and GNN *January 2023 - April 2023*
Incorporated pre-trained LLMs with GNNs for link prediction in knowledge graphs. [Presentation][Code]

Technical Skills

Programming Languages:	Python, C++, C, Java
Data Science Tools:	PyTorch, Claude Models: Sonnet and Opus, LORA for LLM Finetuning, Strands-SDK for LLM Agents, AWS Sagemaker AI, Bedrock Client, NumPy, Pandas, Matplotlib, NetworkX, PyGOD, OpenCV
Web Development:	FrontEnd - HTML, JavaScript, CSS | BackEnd - Flask, MySQL, MongoDB, ReAct
Miscellaneous:	Git, Jupyter Notebook, LaTeX

Courseworks and Problem Solving

- **Related Courses:**
  - Reinforcement Learning, Reasoning with LLMs, Natural Language Processing, Artificial Intelligence, Deep Learning, Data Mining, Statistical Machine Learning, Computation & Machine Learning over Graphs, Algorithm Design, Analysis, and Implementation, Database Systems, Mathematical and Statistical Analysis for Engineers, Linear Algebra, Introduction to Probability and Statistics, Compiling And Programming Systems, Data Communication And Computer Networks
- **Problem Solving**
  - Solved 1000+ problems in different Online Judges including Codeforces, Codechef, and UVA, *Stopstalk Profile*.
  - Reached Expert (Rating 1640) in Codeforces and 5* (Rating 2160) in Codechef, Username: *amitroy7781*.

Teaching Experinence

- Teaching Assistant - Department of Computer Science, Purdue University
  - CS 37300 : Data Mining and Machine Learning, Spring 2024, Fall 2024, Spring 2025, Fall 2025, Spring 2026, Fall 2026
  - CS 57800 : Statistical Machine Learning, Fall 2023

Awards and Achievements

- 2022: **"Ross Fellowship"** by Purdue University for outstanding PhD-track students to graduate programs at Purdue.
- 2022: "**University Grants Commission of Bangladesh Scholarship**" for the first position in the Faculty of Engineering and Technology, University of Dhaka *[Certificate]*
- 2022: **"Professor Dr. M Lutfur Rahman Award"** for securing the highest CGPA in the Bachelor of Science Examination of 2019 in Computer Science and Engineering, University of Dhaka. *[Certificate]*

References

Dr. Bharat K. Bhargava
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Department of Computer Science and Engineering
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